

CS 401: Introduction to Artificial Intelligence

Module 1: AI Foundations and Supervised Learning

Artificial Intelligence (AI) is a branch of computer science focused on creating systems capable of performing tasks that normally require human intelligence, such as reasoning, learning, and language understanding.

Key Branches:

- Machine Learning (ML): Allows computers to learn patterns from data without being explicitly programmed.
- Deep Learning (DL): Utilizes multi-layered neural networks inspired by biological neural structures.

Supervised Learning:

Supervised learning trains models on labeled datasets where inputs map to known target outputs.

Common tasks include:

1. Classification: Categorical prediction (e.g. spam detection, disease diagnosis, image recognition).
2. Regression: Continuous value prediction (e.g. house prices, stock forecasting, temperature estimation).

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Module 2: Unsupervised Learning & Reinforcement Learning

Unsupervised Learning:

Operates on unlabeled data to discover underlying hidden patterns, groupings, and structures.

Core Techniques:

- Clustering: Groups data points by similarity (e.g. K-Means, customer market segmentation).
- Dimensionality Reduction: Compresses feature space while retaining variance (e.g. PCA, t-SNE).

Reinforcement Learning (RL):

An agent learns optimal decision policies through trial-and-error interaction with a dynamic environment.

The agent takes actions, receives environmental state updates, and optimizes for cumulative rewards.

Key applications include robotics, autonomous driving, game playing (AlphaGo), and automated resource scheduling.

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Module 3: Natural Language Processing & Large Language Models

Natural Language Processing (NLP):

Enables computing systems to read, parse, interpret, and generate human natural languages.

Core tasks include sentiment analysis, machine translation, named entity recognition, and question answering.

Large Language Models (LLMs):

Trained on massive web-scale text corpora using Transformer architectures and self-attention mechanisms.

LLMs excel at few-shot reasoning, synthetic text generation, summarization, and domain translation.

Retrieval-Augmented Generation (RAG):

Augments LLM generation by retrieving authoritative knowledge chunks from external vector indexes (such as FAISS).

RAG prevents hallucinations and provides exact source citations for verifiable academic integrity.